

Teacher Answer Key

Mechanical Oscillations – Ex 4: Vertical Elastic Pendulum

1. Equilibrium & Mechanical Energy

a) Elongation Δl_0 at Equilibrium:

At equilibrium position O, the solid (S) is subjected to two forces that balance each other:

- The Weight $\vec{W} = m\vec{g}$ (downward).
- The Tension $\vec{T}$ (upward), with magnitude $T = k\Delta l_0$.

Newton's First Law (Principle of Inertia): $\vec{W} + \vec{T} = \vec{0}$.

Projecting on the vertical axis (downward positive): $mg - k\Delta l_0 = 0$.

Result:
$$\Delta l_0 = \frac{mg}{k}$$

b) Verification of Mechanical Energy (ME):

The Mechanical Energy is the sum of Kinetic (KE), Gravitational Potential (GPE), and Elastic Potential Energy (EPE).

Reference for GPE: Horizontal plane through O ($x = 0$). The axis x is descending.

1. $KE = \frac{1}{2}mv^2$

2. $GPE = -mgx$ (Since the object is at position x below reference).

3. $EPE = \frac{1}{2}k(\text{deformation})^2$. The total elongation is $\Delta l = \Delta l_0 + x$.

$$EPE = \frac{1}{2}k(\Delta l_0 + x)^2 = \frac{1}{2}k(\Delta l_0^2 + 2x\Delta l_0 + x^2)$$

$$EPE = \frac{1}{2}k\Delta l_0^2 + k\Delta l_0x + \frac{1}{2}kx^2$$

Summing them up:

$$ME = \frac{1}{2}mv^2 - mgx + \frac{1}{2}k\Delta l_0^2 + k\Delta l_0x + \frac{1}{2}kx^2$$

Since $mg = k\Delta l_0$ (from part a), the term $-mgx$ cancels with $+k\Delta l_0x$.

Result:
$$ME = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 + \frac{1}{2}k\Delta l_0^2$$

| Note: Since $\frac{1}{2}k\Delta l_0^2$ is constant, the variation of ME depends only on v and x .

2. Friction Neglected

a) Verification of k value:

We apply Conservation of Mechanical Energy between two states:

State A (Maximum displacement): $x = X_m = 4 \text{ cm} = 0.04 \text{ m}$, $v = 0$.

State B (Equilibrium): $x = 0$, $v = V_0 = 0.5 \text{ m/s}$.

$$ME(A) = ME(B)$$

$$\frac{1}{2}kX_m^2 + \frac{1}{2}k\Delta l_0^2 = \frac{1}{2}mV_0^2 + \frac{1}{2}k\Delta l_0^2$$

The constant term $\frac{1}{2}k\Delta l_0^2$ cancels out:

$$kX_m^2 = mV_0^2 \implies k = \frac{mV_0^2}{X_m^2}$$

Numerical Application ($m = 0.05 \text{ kg}$):

$$k = \frac{0.05 \times (0.5)^2}{(0.04)^2} = \frac{0.05 \times 0.25}{0.0016} = \frac{0.0125}{0.0016}$$

Calculation: $1250/160 = 7.8125$

Result: $k \approx 7.8 \text{ N/m}$

b) Differential Equation:

Conservation of Energy: $\frac{dME}{dt} = 0$.

$$\frac{d}{dt} \left(\frac{1}{2}mv^2 + \frac{1}{2}kx^2 + \text{const} \right) = 0$$

$$mvv' + kxx' = 0$$

With $x' = v$ (since RM) and $v' = x''$:

$$v(mx'' + kx) = 0$$

For $v \neq 0$: $mx'' + kx = 0$

Result: $x'' + \frac{k}{m}x = 0$

c) Nature and Period:

This is the equation of Simple Harmonic Motion (SHM).

$$\text{Period } T_0 = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{0.05}{7.8}}$$

$$T_0 \approx 2\pi\sqrt{0.0064} = 2\pi(0.08) \approx 0.5 \text{ s}$$

Result: $T_0 \approx 0.5 \text{ s}$

3. Damped Oscillations (with Plate)

a) Type of Oscillations:

The amplitude decreases over time (it stops after 8 oscillations). These are **damped oscillations** (or pseudo-periodic).

b) Average Power (P_{avg}):

The power required to maintain oscillations is equal to the rate of energy loss.

$$P_{avg} = \frac{|\Delta ME|}{\Delta t}$$

1. Loss in Energy ($|\Delta ME|$):

Initial Mechanical Energy ($t = 0, x = X_m = 0.04 \text{ m}, v = 0$):

$$ME_i = \frac{1}{2}kX_m^2 + \frac{1}{2}k\Delta l_0^2$$

Final Mechanical Energy (Stopped at equilibrium, $x = 0, v = 0$):

$$ME_f = 0 + 0 + \frac{1}{2}k\Delta l_0^2 = \frac{1}{2}k\Delta l_0^2$$

The variation $\Delta ME = ME_f - ME_i$:

$$\Delta ME = \frac{1}{2}k\Delta l_0^2 - \left(\frac{1}{2}kX_m^2 + \frac{1}{2}k\Delta l_0^2 \right) = -\frac{1}{2}kX_m^2$$

Total Energy Lost ($|\Delta ME|$):

$$|\Delta ME| = \frac{1}{2}kX_m^2 = \frac{1}{2}(7.8)(0.04)^2 \approx 0.00624 \text{ J}$$

2. Duration (Δt):

The pendulum makes 8 oscillations.

$$\Delta t = 8 \times T_0 \approx 8 \times 0.5 = 4.0 \text{ s.}$$

3. Calculation:

$$P_{avg} = \frac{0.00624}{4} = 0.00156 \text{ W}$$

Result: **$P_{avg} = 1.56 \text{ mW}$**